

Validity of Theorem 4(b):

x	y	z	$y \cdot z$	$x \cdot (y \cdot z)$	$x \cdot y$	$(x \cdot y) \cdot z$
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	1	0	0	0
1	0	0	0	0	0	0
1	0	1	0	0	0	0
1	1	0	0	0	1	0
1	1	1	1	1	1	1

Validity of Theorem 5(a):

x	y	$x+y$	$(x+y)'$	x'	y'	$x'y'$
0	0	0	1	1	1	1
0	0	0	1	1	0	0
0	1	1	0	1	1	1
0	1	1	0	1	0	0
1	0	1	0	0	1	0
1	0	1	0	0	0	0
1	1	1	0	0	0	0

Validity of Theorem 5(b):

x	y	$x \cdot y$	$(x \cdot y)'$	x'	y'	$(x+y)'$
0	0	0	1	1	1	1
0	0	0	1	1	0	1
0	1	0	1	1	1	1
0	1	0	1	1	0	1
1	0	0	1	0	1	1
1	0	0	1	0	0	1
1	1	1	0	0	0	0

2.3.3 Operator Precedence

- The operator precedence for evaluating Boolean expressions is (1) parentheses, (2) NOT (3) AND, and OR (4)

Example 2-1: Using basic Boolean theorem prove:

$$(a) (x+y)(x+z) = x + yz$$

Solution:

$$\begin{aligned}
 (x+y)(x+z) &= (x+y) \cdot x + (x+y) \cdot z \quad (\text{Postulate 4(a)}) \\
 &= x \cdot (x+y) + z \cdot (x+y) \quad (\text{Postulate 3(b)}) \\
 &= (x \cdot x + x \cdot y) + (z \cdot x + z \cdot y) \quad (\text{Postulate 4(a)}) \\
 &= (x + x \cdot y) + (z \cdot x + z \cdot y) \quad (\text{Theorem 1(b)}) \\
 &= (x + x \cdot y) + (x \cdot z + y \cdot z) \quad (\text{Postulate 3(b)}) \\
 &= x + (x \cdot z + y \cdot z) \quad (\text{Theorem 6(a)}) \\
 &= (x + x \cdot z) + y \cdot z \quad (\text{Theorem 4(a)})
 \end{aligned}$$